RAKESH CHOWDARY POLINA

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PROFILE

Experienced Software Engineer with a proven track record in developing scalable microservices, implementing robust facial recognition pipelines, and delivering cloud-native solutions. Skilled in Python, Java, Spring Boot, Flask, React.js, and AWS, with expertise in containerization, CI/CD, and Agile methodologies. Adept at optimizing performance, enhancing user engagement, and driving operational efficiency in production environments.

EDUCATION

Master of Science in Computer Science **May 2024**
Oklahoma State University, Stillwater, OK **CGPA: 3.909/4**

SKILLS

Languages:	Java, JavaScript, Python, SQL, C, C++, HTML, CSS, Swift
Frameworks:	React JS, Angular JS, Spring boot, Node JS, Hibernate, Spring MVC, React Native
Databases:	MySQL, MongoDB, NoSQL, PostgreSQL, Cassandra
Web Technologies:	RESTful API, Spring JDBC, JPA, OAuth 2, jQuery, JSP, JMS, RabbitMQ
Web Services:	REST, SOAP
Tools/Platforms:	Docker, Git, JIRA, Jenkins, Postman, Ansible, Kubernetes, Apache Kafka, Terraform, Swagger
AWS Cloud Services:	EC2, S3, EKS, RDS, ELB, Cloud Watch, CloudFront
Build Tools:	Maven, Gradle

WORK EXPERIENCE

Research Data Analyst, Oklahoma State University **May 2023 – May 2024**

- Engineered and implemented a robust facial recognition pipeline using Python and InsightFace (AI framework), adhering to modular programming methods and comprehensive documentation procedures. Conducted quantitative/qualitative analysis to achieve 93.6% precision, including exploratory data analysis (EDA) and preprocessing for model optimization.
- Implemented DBSCAN clustering for 182,000+ images, validating results through statistical methods to enhance data reliability. Automated preprocessing workflows using Linux shell scripting for batch processing on Ubuntu servers.
- Designed an intuitive UI for the labeling tool using Flask (backend) and React.js (frontend), following RESTful API design principles. Streamlined user engagement with version-controlled releases via Git, ensuring traceability of feature iterations.
- Designed SQL database schemas with rigorous normalization, documenting ER diagrams and query workflows. Managed user interaction data using PostgreSQL, reducing redundancy by 18% through indexing optimizations.

Software Engineer, Ericsson, India **July 2019 – July 2022**

- Architected scalable Java/Spring Boot web applications using RESTful APIs, applying OOP principles and documenting architectural decisions. Improved performance by 25% via reactive programming with Spring WebFlux and load testing using JMeter.
- Optimized PostgreSQL databases with Spring Data JPA, reducing query latency by 20%. Authored schema documentation and stored procedure guidelines for team alignment.
- Built React.js interfaces with Redux for state management, adhering to component-driven development. Integrated RESTful APIs using Axios, improving responsiveness by 30% through lazy loading.
- Implemented CI/CD pipelines with GitHub Actions, automating deployments to Linux environments (RHEL). Reduced deployment time by 35% using Docker containers and Kubernetes orchestration.
- Secured APIs using Spring Security with OAuth2, documenting protocols via Swagger. Containerized legacy systems using Docker, achieving 99.8% uptime on AWS EKS clusters.
- Conducted code reviews using Git in Agile sprints, mentoring junior developers on Spring/React best practices. Resolved production issues using Linux diagnostic tools (e.g., journald, syslog).
- Applied software engineering techniques including unit testing (JUnit/Mockito) and integration testing (Postman), reducing post-deployment defects by 25%.
- Followed AGILE (SCRUM) methodologies with bi-weekly sprint planning and daily status meetings to monitor project progress and ensure alignment with product development goals. Collaborated closely with cross-functional teams, including development, operations, and quality assurance, to ensure seamless communication and coordination.

PROJECTS

Sleep data analysis - Applied LSTM network and Granger causality techniques to investigate sleep data on a dataset comprising over 5,000 individuals' sleep patterns, unveiling insights into temporal dynamics and causal connections within the dataset

HireOSUgrads App - Developed an iOS application for university students and recruiters, featuring intuitive swipe gestures for seamless interactions. Utilized Swift, followed MVC architecture and integrated SQLite for robust data management